Satyam Awasthi

Researcher & Software Engineer

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Research Summary

Computer Science researcher working on **neural interfaces**, **physiological sensing**, and **uncertainty-aware control**. Current work spans **EEG-based brain–computer interfaces** and **robust output-feedback model predictive control**.

Research Interests

Neural Interfaces, Brain–Computer Interfaces, Physiological & Wearable Sensing, Human–Machine Interaction, Machine Learning for Biosignals, Robust State Estimation, and Model Predictive Control.

Research Experience

Autonomous Systems Group, Imperial College London — Jul 2026 – Present
Research Collaboration — Advisor: Dr. Johannes Köhler

- Investigating **state-dependent estimation and tracking-error bounds** for nonlinear robust output-feedback MPC with state-dependent process and measurement uncertainty.
- Developing coupled observer and tracking-error tube propagation using **contraction-based certificates** for robust constraint tightening and uncertainty-aware trajectory planning.

Four Eyes Laboratory, University of California, Santa Barbara — Sep 2021 – Jun 2023
Graduate Researcher — Advisor: Prof. Tobias Höllerer; Co-Advisor: Prof. Michael Beyeler

- Developed **EyeTTS**, an open-source framework for evaluating and calibrating eye tracking during mixed-reality locomotion, resulting in first-author publications at IEEE ISMAR Adjunct and IEEE VR Workshops.
- Designed human-subject experiments and analyzed **gaze-tracking accuracy, calibration drift, and motion-dependent sensing error** across multiple mixed-reality devices.
- Released a reproducible **calibration framework** and **multi-device eye-tracking dataset** containing gaze trajectories, calibration measurements, and locomotion data.

Autonomous Ground Vehicle & Swarm Robotics Group, IIT Kharagpur — Jan 2016 – Dec 2018
Undergraduate Researcher

- Contributed to **Eklavya 6.0**, IIT Kharagpur's autonomous ground vehicle that finished **Runner-Up** at the **Intelligent Ground Vehicle Competition (IGVC) 2018**.
- Developed perception and navigation components integrating LiDAR, IMU, GPS, vision-based perception, state estimation, and motion planning.
- Worked on decentralized multi-agent coordination using relative-pose estimation and communication over an ad-hoc mesh network.

Publications

Peer-Reviewed Conference Papers

SmartWatch for Wall Writing: Real-time Transcription of Wall Writing from Inertial Sensing
Snigdha Das, **Satyam Awasthi**, Abdul Shamnar P, Pradipta De, Sandip Chakraborty, and Bivas Mitra.
COMSNETS, 2022.
DOI: 10.1109/COMSNETS53615.2022.9668531

Workshop & Poster Publications

Eye Tracking Performance in Mobile Mixed Reality
Satyam Awasthi, Vivian Ross, Sydney Lim, Michael Beyeler, and Tobias Höllerer.
IEEE VR Workshops (VRW), 2024.
DOI: 10.1109/VRW62533.2024.00321

EyeTTS: Evaluating and Calibrating Eye Tracking for Mixed-Reality Locomotion
Satyam Awasthi, Vivian Ross, Michael Beyeler, and Tobias Höllerer.
IEEE ISMAR Adjunct, 2023.
DOI: 10.1109/ISMAR-Adjunct60411.2023.00104

Selected Research Projects

EEG-Based BCI for 3D Navigation 2026 – Present
Independent Neural Interface Project

- Developing a **4-channel EEG acquisition and decoding pipeline** for real-time control of an interactive 3D navigation environment.
- Building an initial **SSVEP-based intent-decoding system** using neural-signal preprocessing and spectral / correlation-based features for discrete movement commands.
- Investigating confidence thresholds and temporal evidence accumulation to characterize the trade-off between **decoding reliability and command latency**.

SmartWatch for Wall Writing 2019 – 2022
Wearable Inertial Sensing for Human–Computer Interaction

- Contributed to a wearable sensing system for **real-time transcription of free-form wall writing** from smartwatch inertial measurements.
- Explored the use of human wrist-motion signals as an unobtrusive input modality for recognizing physical writing gestures.

JitterNot Jan 2022 – Mar 2022
Learning-Based Model Predictive Control for Adaptive Video Streaming (UCSB)

- Developed an LSTM-based predictor for network throughput to support closed-loop decisions under non-stationary network dynamics.
- Designed a receding-horizon **model predictive control (MPC)** framework for adaptive bitrate selection under bandwidth constraints.

In Motion — Wearable Human Activity Recognition Nov 2019
DeepConvLSTM Modeling of Inertial Sensor Signals (IIT Kharagpur)

- Designed a streaming inference pipeline for **high-frequency IMU signals** enabling temporal human-activity classification.
- Implemented a DeepConvLSTM architecture combining convolutional feature extraction with recurrent temporal modeling of wearable sensor sequences.

Education

University of California, Santa Barbara (UCSB) Santa Barbara, CA
M.S. in Computer Science (GPA: 3.91/4.00) Sep 2021 – Jun 2023

Indian Institute of Technology, Kharagpur (IIT Kharagpur) Kharagpur, India
B.Tech. in Electrical Engineering, Minor in Computer Science (GPA: 8.98/10.0) Jul 2016 – Jul 2020

Industry Experience

Intuit Inc Mountain View, CA · Mar 2024 – Jan 2026
Software Engineer 2

- Built dual-stage agent pipelines for intent routing, structured tool execution, and generative user-interface composition.
- Developed production platform infrastructure supporting large-scale client applications and internal developer workflows.

Yahoo Inc Mountain View, CA · Apr 2023 – Nov 2023
Software Dev Engineer II (IC3)

- Developed production Android systems for Yahoo Mail, including data-driven inbox commerce and expense-management features.

Coupang Global LLC Mountain View, CA · Jun 2022 – Sep 2022
Software Engineering Intern

- Developed user-facing search and product-discovery features for Coupang's Android commerce platform.

Samsung Research Institute Noida, India · Sep 2020 – Sep 2021
Software Engineer (CL 2)

- Developed Android platform and device applications involving media processing, privacy, and isolated user-storage architectures.

Teaching

University of California, Santa Barbara Santa Barbara, CA
Graduate Teaching Assistant Sep 2021 – Jun 2023

- Led instructional labs, developed programming assignments, and supported senior capstone teams across CS189 (Capstone), CS184 (Android), CS24 (C++ Data Structures), and CS16 (Problem Solving).

Technical Skills

Programming: Python, C++, MATLAB, Java, Kotlin

Machine Learning & Signal Processing: PyTorch, NumPy/SciPy, OpenCV, Time-Series Modeling, Wearable/Inertial Signal Processing

Control & Estimation: Model Predictive Control (MPC), Robust MPC, Contraction-Based Methods, State Estimation (EKF), Sensor Fusion, Motion Planning

Robotics & Interactive Systems: ROS/ROS2, Gazebo, RViz, Unity, OpenGL

Developer Tools: Git

Academic Service & Peer Review

Peer Reviewer, UIST 2024
ACM Symposium on User Interface Software and Technology

- Reviewed submissions on interactive systems, spatial tracking, and interface hardware.